EUROPEAN SOCIETY OF RADIOLOGY
European Radiology

INVITED REVIEW Open Access

Evolving role of cryoablation and cryoneurolysis for interventional pain management

Daniel M. Düx[1], Dimitrios Filippiadis[2], Frank Wacker[1] and Kristina Imeen Ringe[1*]

Abstract

Despite advances in pain management, pain remains a major global health challenge in both oncological and non-oncological patients. Long-term pharmacological treatment is often restricted by insufficient therapeutic benefit and adverse events, highlighting the need for alternative approaches such as image-guided pain therapies. The idea of treating pain with cold has been recognized for centuries, and today cryotherapy has emerged as a valuable minimally invasive option for pain management. Cryoablation refers to the destruction of tissue with extreme cold and has an increasing role, especially in the treatment of benign (e.g., osteoid osteoma), malignant bone (i.e., metastases), and soft tissue tumors, either with a curative or palliative intent. Cryoneurolysis is a related but distinct application, focusing either on peripheral nerve modulation or nerve destruction. Cryoneurolysis may be highly effective for the treatment of acute or chronic pain, including sympathetic neurolysis in cancer and non-cancer-related pain syndromes. The general approach in patients with tumor-related pain depends on whether the lesion itself can be ablated or not. In non-oncologic patients, cryoneurolysis of the responsible nerve is typically performed following a positive response to a diagnostic nerve block. Potential indications for cryotherapy are constantly evolving; however, evidence, especially for cryoneurolysis, is still limited and based mostly on single-center case series and retrospective studies. Standardized ablation protocols are lacking, and for the duration of cycles, most experts rely on imaging visualization and validated protocols. Equally important, successful integration of cryoablation and cryoneurolysis into daily practice requires structured workflows across different patient pathways and interdisciplinary management.

Key Points

Question *Pain management remains a major global health challenge and an unmet need in oncological and non-oncological patients.*

Findings *Cryotherapy, in terms of cryoablation and cryoneurolysis, is increasingly incorporated in multimodal curative and palliative pain care concepts, for treatment of acute and chronic pain syndromes.*

Clinical relevance *Advancement and validation of standardized treatment protocols are needed to strengthen evidence and further enhance clinical implementation in interventional radiology practice and interdisciplinary patient care.*

Keywords Cryoablation, Cryoanalgesia, Cryoneurolysis, Pain management

*Correspondence:
Kristina Imeen Ringe
ringe.kristina@mh-hannover.de
[1]Department of Diagnostic and Interventional Radiology, Hannover Medical School, Hannover, Germany
[2]2nd Department of Clinical Radiology, University General Hospital "ATTIKON", Haidari, Greece

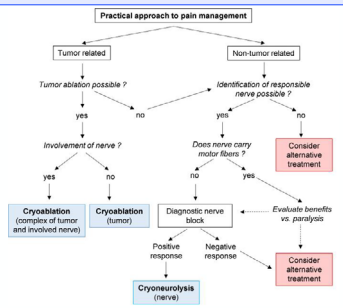
Springer

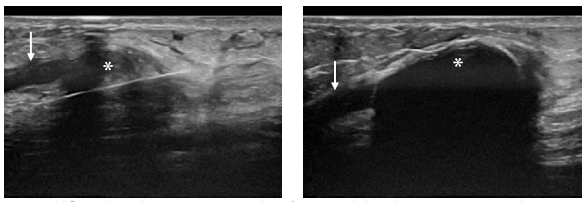
© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

Graphical Abstract

Evolving role of cryoablation and cryoneurolysis for interventional pain management

How can image-guided cryotherapy contribute to acute and chronic pain management?





US-guided cryoneurolysis of a sensitized post-amputation stump neuroma (*) of the cutaneous brachii medialis nerve (arrow)

Advancement and validation of standardized treatment protocols is needed to strengthen evidence and enhance implementation in interdisciplinary patient care.

Eur Radiol (2026) Düx DM, Filippiadis D, Wacker F, Ringe KI; DOI: 10.1007/s00330-026-13035-z

EUROPEAN SOCIETY OF RADIOLOGY European Radiology

Introduction

Despite advances in pain management, pain remains one of the most prevalent and debilitating symptoms in both oncologic and non-oncologic patient populations, representing a major global health burden and a key driver of healthcare utilization. Chronic pain, defined as persistent or recurrent pain lasting longer than 3 months [1], is prevalent in approximately 51–66% of cancer patients, with the majority of patients remaining undertreated [2]. Similarly, chronic non-cancer pain, present in approximately 20–30% of adults worldwide, is associated with impaired physical function, mental health problems, and a diminished quality of life [3]. Traditional pain treatment concepts rely on pharmacological therapy, with or without complementary physiotherapy, radiation, or even invasive surgery [4]. The reliance on long-term pharmacological management, particularly opioids, is associated with additional risks, including dependency, tolerance, and side effects. In this context, image-guided interventional techniques have emerged as an important adjunct or alternative to conventional approaches and are increasingly being incorporated into multidisciplinary treatment concepts.

The idea of treating pain with cold has been recognized for centuries and dates back to the ancient Greeks, Persians, and Egyptians, applying snow to wounds for pain relief [5]. In 1849, James Arnott described the use of severe cold to treat certain types of cancer-related and neuropathic pain [6]. A milestone for modern cryotherapy has been the landmark publication by Lloyd et al in 1976, who used a hollow probe that delivered nitrogen to peripheral nerves in 64 patients with intractable pain, coining the term “cryoanalgesia” [7]. Throughout the years, with the availability of liquid nitrogen, the introduction and downsizing of vacuum-isolated probes, cryotherapy has evolved into the technique as we know it today. In conjunction with image-guidance, it enables safe and precise targeting of superficial and deep structures for treatment of neoplastic and non-neoplastic pain [8].

Of note, the armamentarium for interventional pain therapies is constantly evolving, including ablation (thermal and non-thermal), bone consolidation procedures (vertebral augmentation, percutaneous osteosynthesis), intra-arterial therapies, and more cutting-edge techniques like electrochemotherapy [9]. With regards to thermal ablation, heat-based techniques in terms of microwave ablation (MWA) and radiofrequency ablation (RFA) are very well established. Specifically pulsed RFA is broadly applied with excellent results [10]. With cryotherapy having experienced a revival in recent years, in this review,

we will focus on the evolving role of cryoablation and cryoneurolysis, mechanisms of action, clinical applications, and evidence, in cancer-related and non-cancer-related pain syndromes.

Techniques and mechanisms of action

The terms cryoablation, cryoanalgesia, cryoneuroablation, and cryoneurolysis are frequently used interchangeably to describe cryotherapy. Even though the concept of employing low temperatures is common in these procedures, they are distinct interventions marked by different operating temperatures and mechanisms of action.

Cryoablation

Cryoablation refers to the process of destroying tissue (e.g., tumors) with extreme cold [11]. Tumor cell viability sharply decreases with declining temperature. At temperature ranges between 0 °C and −20 °C, cancer cell survival is possible. The majority of cells die as temperatures approach −40 °C [12]. It has been shown that some cancer cells survive at even lower temperatures, necessitating repetitive exposition to temperatures below −40 °C in order to ensure cell death [13, 14]. For cryoablation, one or more cryoprobes (ranging from 8 to 17 gauge in diameter) are inserted under image guidance into the target tissue. Available devices generate extreme cold either via the use of compressed gases (Joule-Thomson effect) or employing liquid nitrogen, the latter achieving its cooling effect when transformed from liquid to gas by the process of boiling [15]. In systems that make use of the Joule-Thompson effect, cryoprobes consist of a hollow tube with two chambers, which are flushed with compressed gases. The gas is released at the probe tip through the smaller inner chamber. Rapid decompression results in the supercooling of most gases, such as argon (Joule-Thomson effect), and gradually freezing of the surrounding tissues, forming an ice ball. The gas is then vented back into the probe through the larger outer chamber, making sure that no gas escapes into the patient's tissues from this closed system [5].

With current cryoablation systems, typically probe tip temperatures up to −140 °C (argon gas systems) or up to −196 °C (liquid nitrogen systems) can be achieved. Cryoablation causes cellular death and necrosis of tissues by direct mechanisms, which cause cold-induced injury to cells, and indirect mechanisms, inducing changes to the cellular microenvironment and impair tissue viability [15]. Alternating cycles of tissue freezing and thawing within the ice ball cause inter- and intracellular ice crystal formation, leading to cell dehydration, cellular membrane disruption, and complete breakdown of cellular structures [16]. Secondary hypoxia and tissue necrosis are achieved as thrombi form within the local circulation. Active thawing is realized via the release of helium into the probe (Joule-Thomson effect) or electric heating of the probe. The inflow of water during thawing increases cell volume, enhancing cell membrane disruption at subsequent freezing. Repetition of the freeze-thaw cycle subjects the tissue again to the injurious elements and is considered to be important in order to ensure proper destruction of the target lesion [17].

In cryoablation, cell death occurs in the form of necrosis and apoptosis. It has been hypothesized that intracellular contents of cancer cells stay intact, allowing the immune system to induce an immune-specific reaction, including the release of tumor antigens that are phagocytosed by dendritic cells. Disturbance of mitochondrial activities due to apoptosis causes the initiation of Bax proteins, further triggering downstream apoptotic pathways [18, 19]. This specific reaction can, in theory, affect cancer cells outside the ablated tissue. It has been observed that this so-called abscopal effect may be even enhanced by combination with immunotherapy. However, as this immune response mechanism is not yet fully understood, it is a matter of ongoing research [18].

Cryoneurolysis

Cryoneurolysis, also referred to as cryoanalgesia or cryoneuroablation, has emerged as a related but distinct application focused either on peripheral nerve modulation or nerve destruction, depending on the temperature reached. Studies have shown that cryoneurolysis with temperatures between −20 °C and −100 °C induces a predictable histopathological sequence including transduction disruption, Wallerian degeneration, and axonal regeneration along an intact connective tissue scaffold (axonotmesis) [20–22]. Temperatures below −100 °C cause nerve injuries with risks of permanent nerve damage (i.e., neurotmesis). These temperature thresholds are derived from experimental studies [21, 22], and translation to clinical practice is not uniform in the present literature. For peripheral nerve modulation, the objective of treatment is a controlled Sunderland type II injury (axonotmesis) with a preserved endoneurial sheath, which permits predictable regeneration and minimizes neuroma formation [20]. To ensure consistent, reversible Sunderland type II injury without risk of overtreatment, a recent CIRSE (Cardiovascular and Interventional Radiology of Europe) expert panel recommends temperatures between −20 °C and −80 °C [8]. Pain conduction is interrupted for weeks to months, with axonal regeneration occurring at a rate of approximately 1–2 mm per day [23]. For cryoneurolysis, either tumor ablation probes or dedicated nerve analgesia probes (14–24 gauge) are employed. The latter make use of carbon dioxide or nitrous oxide for ice ball formation and are characterized

by integrated neurostimulators to assess sensory and motor responses for precise targeting, at the expense, however, of smaller ice ball formation [5, 8].

Practical approach and considerations

The herein presented practical considerations are based on a recent consensus paper on the application of cryoneurolysis, published by a CIRSE expert panel [8]. In this review, we will address key aspects in brief, but we would also like to draw the reader's attention to the mentioned publication for further reference. Integrating cryoablation and cryoneurolysis into daily practice requires structured workflows across outpatient clinics, interventional radiology suites, and perioperative care pathways. Especially in oncological patients as well as in patients with complex neuropathic pain syndromes, indications should be discussed interdisciplinary. Appropriate patient selection and education (also with regard to outcome expectations) are essential steps to guiding treatment success. Organizational models for a patient-centered approach include dedicated interventional pain centers, multidisciplinary clinical (tumor) boards, and acute transitional pain services. In this context, the collaboration between anesthesiologists and interventional radiologists yields strong synergistic effects, combining precise image-guided procedures and in-depth expertise in pharmacological personalized medicine.

Cryoablation and cryoneurolysis have distinct advantages compared with other thermal and non-thermal ablation techniques. Real-time visualization of the ice ball at the probe tip enables safe and controlled treatment, especially when targeting lesions near critical structures. The ice ball can be shaped individually according to the desired size and configuration. Besides the number of probes used, the length of the ice ball depends on the size of the active tip (and the diameter of the probe) whereby the diameter of the ice ball can be adjusted by modifying the flow rate of the gas [11]. In this context, it is important to keep in mind that the border of the ice ball as visible on imaging corresponds to the 0 °C isotherm, which is not lethal. Reliable cell destruction requires temperatures of at least −40 °C combined with repeated freeze-thaw cycles. Especially when compared with heat-based ablation, cryoablation has an intrinsic analgesic effect, often allowing the procedure to be performed under conscious sedation or local anesthesia rather than general anesthesia. In fact, local anesthesia or conscious sedation is preferred, especially for cryoneurolysis, facilitating real-time physiologic assessment of sensory and motor function, while reserving general anesthesia for cancer patients or complex cases [8].

Probes are inserted under image-guidance (ultrasound, CT, or MRI), enabling safe and precise targeting of superficial and deep structures. If necessary, adjunct protective techniques are applied to prevent non-target ablation, including, for example, hydrodissection (saline), balloon displacement, or gas insufflation [8]. Skin protection for superficial targets is important and may be achieved, for example, by applying heat pads or sterile gloves filled with warm water. The choice of image-guidance depends primarily on target visibility, availability, and operator preference. The general approach in patients with tumor-related pain depends on whether the lesion itself can be ablated or not (Fig. 1). If tumor ablation is not feasible, cryoneurolysis should be considered if a responsible nerve can be identified [24]. In non-oncologic patients, cryoneurolysis of the responsible nerve is performed following a positive response to a diagnostic nerve block [24] (Fig. 1).

Prior to treatment, comprehensive informed patient consent is required, including risks, contraindications, and especially expectations with regard to duration of pain relief and nerve regeneration [8]. Contraindications include anatomic barriers, non-correctable bleeding disorders, active infection at the treatment site, cold urticaria, cryoglobulinemia, and Raynaud's syndrome [8]. With regards to bleeding risk stratification, we would like to draw the readers' attention to the current CIRSE SOP on peri-anticoagulation management [25]. Treatment should be performed under meticulous sterile conditions. On the use of routine prophylactic antibiotics, there is currently no consensus. It should be considered especially in patients who are immunocompromised, have diabetes, prosthetic devices, or a history of prior radiation, as well as in cases requiring large-bore probes (> 14 gauge) with deep access [8].

Follow-up imaging after cryoablation or cryoneurolysis (apart from the purpose of tumor staging) is reserved for suspected complications or unexpected pain recurrence. A structured clinical follow-up, however, is recommended to evaluate treatment response, characterize benefit duration, and guide re-treatment, if needed. The use of standardized patient-reported outcome measures (PROMs), such as the visual analog scale (VAS) or numeric rating scale (NRS), is highly advisable [26]. In the context of cryoneurolysis, the CIRSE expert panel, for example, recommends clinical follow-up at 1 week, 4 weeks, 3, 6, and 12 months [8].

Cryoablation for pain management

There is increasing evidence that lesion-directed cryoablation targeting a painful mass may provide effective pain relief (Table 1). In this context, cryoablation has an increasing role in the management of bone and soft tissue tumors [27]. Ablation itself can be performed with a curative or palliative intent, depending on the underlying

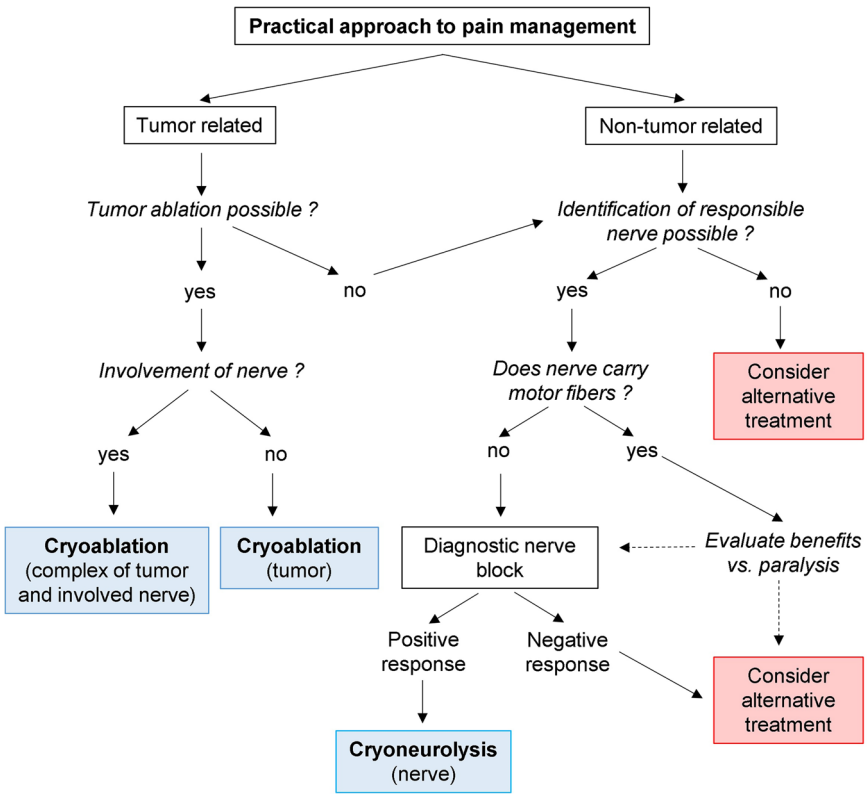


Fig. 1 Practical approach to treatment of acute or chronic pain in patients by means of cryoablation and cryoneurolysis (modified according to [24])

condition and indication. Of note, there are currently no standardized ablation protocols. For malignant lesions, double freeze-thaw cycles ranging between 5 and 10 min are most commonly applied. With the primary aim of pain palliation, and if treatment of the entire lesion is not feasible, cryoprobes are placed in a manner that the resulting ice ball includes especially the lesion-bone interface (Fig. 2). In addition, tumor debulking may decompress adjacent nerves responsible for radiating pain, thereby providing symptomatic relief.

Bone metastases

The palliative treatment of bone metastases has become one of the main indications for percutaneous cryoablation [27]. A recent systematic review including 15 studies with 376 patients showed that cryoablation for palliation of painful bone metastases resulted in significant pain reduction from day 1 up to 6 months after the procedure (maximum mean visual analog scale (VAS) difference 5.8 points at 4 weeks) along with an improved quality of life and a decreased need for analgesics [28]. The reported overall rate of minor and major complications was 12.7%, with the most frequent complications including fever (3.9%), fractures (2.3%), and plexus injuries (1.8%) [28]. Compared to radiofrequency ablation (RFA), it has been shown that cryoablation may yield greater early pain relief, associated with shorter hospital stays (mean 1.0 vs. 3.5 days), while RFA appears to offer higher medium- to long-term benefits [29, 30]. As treatment options for patients with painful metastases evolve, combination therapy may be an expedient approach [11]. Combining cryoablation with bone consolidation when treating lesions in weight-bearing locations may not only prevent skeletal-related events (bone pain, pathologic fracture, neurologic compression) but also provide synergetic pain palliation and tumor control [27, 31, 32]. Similarly, it has been shown that combining ablation with radiation may provide even superior pain control compared with either treatment alone [33].

Osteoid osteoma

Osteoid osteoma (OO) is a benign bone-forming lesion, primarily affecting children and young adults with a male predominance. It comprises approximately 10% of all bone tumors and is characterized histologically by a central nidus made of osteoblasts and osteoid with peripheral nerve and vascular supply. The excessive production of prostaglandins in the nidus leads to local inflammation,

Table 1 Most important studies summarizing clinical evidence for cryoablation of bone lesions

Target and clinical indication	Study design/description	Main findings	Reference
Bone metastases			
Osteolytic bone metastases or associated periosteal reaction with soft tissue mass (various locations)	Retrospective single-center study including 58 patients undergoing cryoablation ($n = 36$) or RFA ($n = 22$); lesion size ranging from 1.9 to 11.6 cm	Cryoablation was associated with a greater reduction in analgesic dose (morphine-equivalent dosage; median −23.6 vs. 21.6) in the first 24 h and shorter hospital stays after the procedure (median 1 vs. 3.5 days)	Thacker et al 2011 [30]
Osteolytic or mixed bone metastases (various locations)	Prospective multicenter study including 61 patients (69 lesions, ranging from 1 to 11 cm in size)	Mean worst pain score before ablation was 7.1/10 and decreased to 5.1/10, 4/10, 3.6/10, and 1.4/10 at 1, 4, 8, and 24 weeks after treatment; one major complication occurred (osteomyelitis at the site of ablation)	Callstrom et al 2013 [76]
Osteolytic and osteosclerotic bone metastases treated with curative intent (extremities and spine)	Retrospective single-center study including 89 patients (122 metastases, lesion size ranging from < 2 cm to > 3 cm); patients were treated with either RFA ($n = 74$) or cryoablation ($n = 48$); in 70 lesions, ablation was combined with cementoplasty	Median follow-up 22.8 months; favorable prognostic factors for complete local treatment were oligometastatic status, metachronous and small-sized bone metastases, without cortical bone erosion or neurological structures in the vicinity; choice of ablation technique had no impact on treatment failure	Deschamps et al 2014 [77]
Solitary osteolytic and mixed painful bone metastases (various locations)	Retrospective single-center study including 175 patients (propensity matched analysis of subgroups included 25 patients who underwent radiation 15 days after cryoablation, 125 patients treated by radiotherapy alone, and 25 patients treated only using cryoablation); tumor size ranging from 3.4–6 cm	A higher proportion of patients treated with cryoablation (32%) or cryoablation followed by radiotherapy (72%) experienced a complete response compared with patients treated by radiotherapy alone (11.2%); additional radiotherapy significantly improved the rate of complete response compared with cryoablation alone, paralleling with an improved self-rated quality of life	Di Staso et al2015 [33]
Mixed painful bone metastases (various locations)	Systematic review including 15 studies (376 patients)	All studies reported significant pain reduction between 1 day and 6 months after cryoablation; the highest mean difference between pre- and post-procedure scores was 5.8 (on the VAS) after 4 weeks; the overall rate of complications was 12.74%; cryoablation improved quality of life in cancer patients and decreased the need for analgesics	Khanmohammadi et al 2023 [28]
Mixed painful bone metastases (various locations)	Single-arm meta-analysis of 25 studies comparing cryoablation and RFA	RFA and cryoablation significantly reduced pain at all follow-up timepoints; RFA had the greatest pain reduction at 6 months; cryoablation had a greater effect at 24 h but a smaller effect at 6 months; adverse events were mostly mild with both techniques	Alaqeel et al 2025 [29]
Osteoid osteoma			
Osteoid osteoma (spine, extremities); adult patients	Retrospective single-center study including 21 patients	Technical success 100%; median pain on the VAS was 8/10 before the procedure and 0/10 at primary (6 weeks) and secondary follow-up (between 6 and 40 months); three minor	Santiago et al 2018 [43]

Table 1 continued

Target and clinical indication	Study design/description	Main findings	Reference
		complications (one mild skin burn and two cases of transient soft tissue swelling and mechanical pain); one patient reported symptom recurrence at secondary follow-up and underwent a successful re-cryoablation procedure	
Osteoid osteoma (various locations); pediatric and adult patients	Systematic review and analysis including 36 studies (1863 ablations in 1798 patients)	Overall success rate 91.9%; technical failure, clinical failure, recurrence, and complication rates were 0.3%, 2.1% 5.6% and 2.5% without a significant difference between RFA and cryoablation ($p = 0.92$)	Lindquester et al 2020 [38]
Osteoid osteoma (spine, extremities); adult patients	Retrospective single-center study including 50 patients; in 60% nidus was targeted from an extraosseous position without penetration of bone	Overall clinical success rate 96%; median pain on VAS 8 before ablation and 0 at primary (6 weeks) and secondary (18–90 months) follow-up; 3 patients had minor complications ($n = 2$ soft tissue swelling; $n = 1$ mild skin burn)	Le Corroller et al 2021 [40]
Osteoid osteoma (various locations); pediatric and adult patients	Systematic review and meta-analysis including 45 studies (1528 patients); inclusion of all thermal ablation techniques (RFA $n = 1150$; cryoablation $n = 75$; laser $n = 276$; MWA $n = 44$)	All ablation methods resulted in significant immediate and lasting pain reduction; technical and clinical success rates and major complications did not differ between ablation techniques; overall recurrence of pain occurred in 4.06% at an average of 11 months post-procedure	Shanmugas-undaram et al 2021 [39]
Osteoid osteoma (various locations); pediatric and adult patients	Retrospective single-center study including 93 patients; comparison of RFA (57 treatments) and cryoablation (42 treatments)	Overall clinical success 95% (RFA: 93%; cryoablation: 97%; $p = 0.343$); pain recurred in 7% (RFA group) and 3% (cryoablation group; $p = 0.202$); in cryoablation, one major (osteomyelitis) and one moderate adverse event (neuropathic pain) were reported	Thompson et al 2026 [78]

RFA radiofrequency ablation, *VAS* visual analog scale, *MWA* microwave ablation

hyperemia, and abnormal bone growth. Patients typically suffer from (nocturnal) pain, which is responsive to non-steroidal anti-inflammatory drugs [34, 35]. Today, percutaneous thermal ablation with complete destruction of the nidus is the treatment of choice. Ablation of osteoid osteoma was first described using RFA in 1992 [36]. Since then, there has been a continuously enlarging body of evidence that percutaneous thermal ablation (including cryoablation and MWA) of osteoid osteoma is safe and effective [37–39]. In addition to the already mentioned advantages, cryoablation has another distinct benefit, which is particularly useful when treating bone-forming tumors such as osteoid osteoma. Opposed to the inductive nature of RFA heat generation, the conductive nature of ice formation allows the ice ball to be transmitted across sclerotic bone (similar to MWA), which is less affected by tissue impedance as compared to RFA [38], enabling treatment of the lesion from an extraosseous position (en-passant approach) [40] (Fig. 3). Again, there are no standardized ablation protocols. While some authors propagate double freeze-thaw cycles [40], current evidence and practice support that a single freeze (5–8 min) followed by passive thawing without track ablation is sufficient in benign lesions [41]. Studies, mostly retrospective in nature, have demonstrated a primary success rate of 96–100% for cryoablation of osteoid osteoma without any major complications [40, 42]. A recent meta-analysis comparing RFA and cryoablation, including 1798 patients, depicted no significant differences in overall treatment success (91.1%) [38]. However, one potential benefit of cryoablation is the reduction of immediate post-procedural pain when compared to heat-based ablation techniques [43].

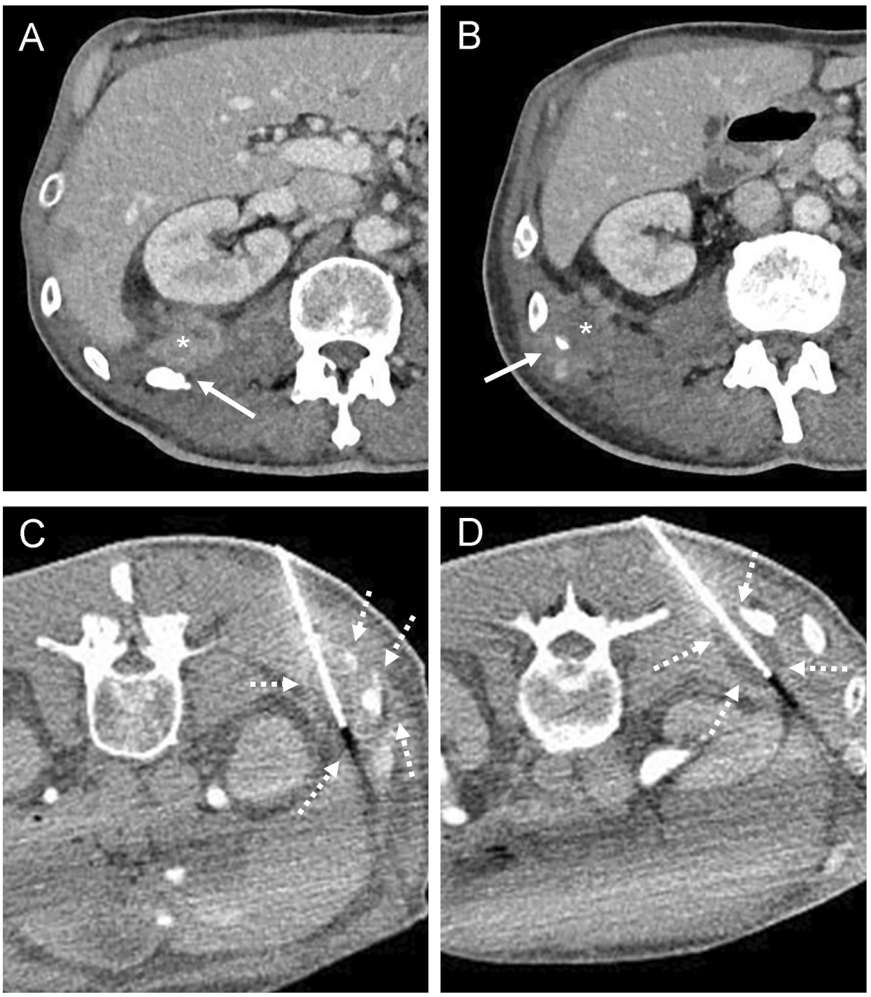


Fig. 2 Palliative cryoablation of a painful abdominal wall metastasis. 53-year-old male patient with metastasized urothelial carcinoma. Pre-interventional CT identified a retroperitoneal metastasis (* in **A** and **B**) infiltrating the abdominal wall, including the 11th and 12th rib (arrows in **A** and **B**), causing intractable pain. Cryoablation was performed under CT guidance with the patient in the prone position (**C**, **D**). Two cryoprobes were placed, making sure that the resulting ice ball included the lesion-bone interface (dashed arrows in **C** and **D**) for maximum pain relief

Evolving indications

Cryoablation is increasingly being used for the treatment of a variety of benign bone and soft tissue tumors [27]. Full discussion of all potential indications for cryoablation would exceed the scope of this manuscript. Osteoblastoma, often referred to as the "big brother" of osteoid osteoma, has traditionally been treated using surgery. Cryoablation has been shown to be very effective with a high safety profile [44]. Emerging indications also in the context of pain palliation include, for example, treatment of chondroblastoma [45], chondromyxoid fibroma [46], and vascular malformations [47]. Even though current evidence for these entities is mostly based on smaller case series, reported outcomes are very encouraging. Similarly, desmoid fibrosis [48, 49] and abdominal wall endometriosis [50, 51] are rapidly evolving indications for cryoablation. Cryoablation is increasingly being incorporated into multimodal treatment concepts of patients with recurrent malignant bone and soft tissue lesions (e.g., sarcoma), especially after failure of conventional treatment techniques, tumor recurrence, or insufficient pain relief [52].

Cryoneurolysis for pain management

In cryoneurolysis, superficial or deep neural structures responsible for neoplastic or non-neoplastic pain are targeted. Probes are placed in the target region under image-guidance with minimal trauma, so the predicted ablation zone will encompass the neural structure of interest [53]. In order to avoid direct mechanical damage

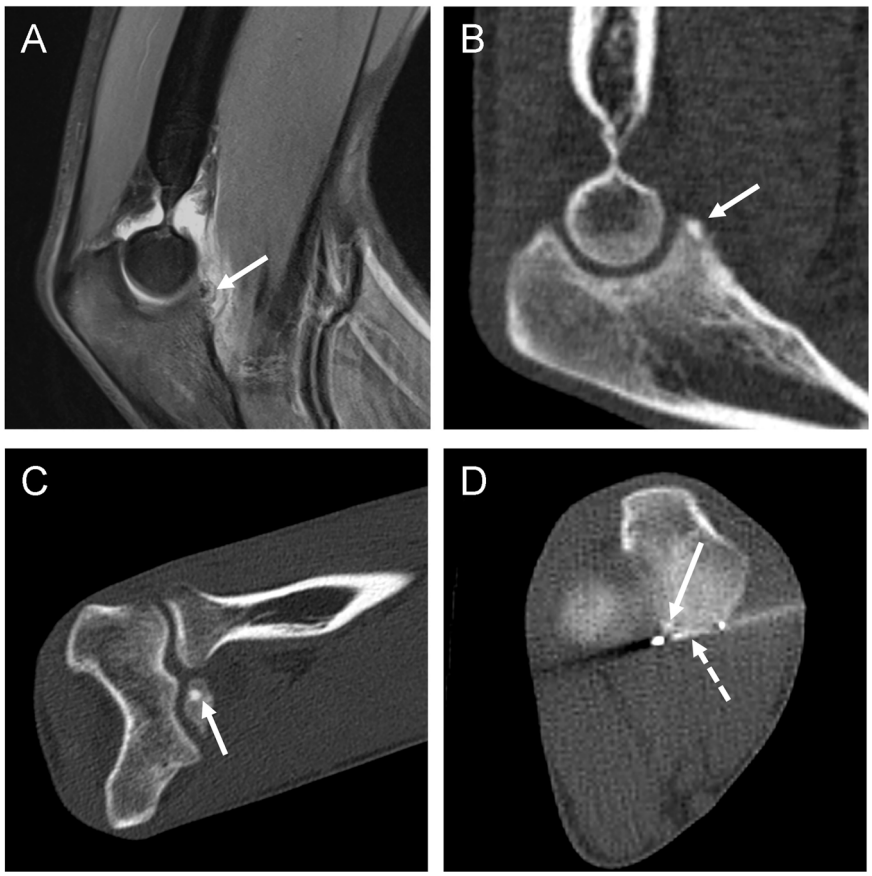


Fig. 3 Cryoablation of an osteoid osteoma using en-passant technique. The cryoprobe is placed adjacent to the cortex, parallel to the nidus, without penetrating it. 22-year-old male patient with nocturnal pain responsive to non-steroidal anti-inflammatory drugs. Sagittal PD-weighted MRI (**A**), sagittal (**B**), and coronal reformatted CT (**C**) depict a 3-mm osteoid osteoma (arrows in **A**–**D**) in the coronoid process. A single cryoprobe was placed under CT-guidance adjacent to the cortex (dashed arrow in **D**) and ablation was performed (single 5-min freeze followed by 3-min passive thaw). At clinical follow-up the next day, the patient was free of pain

to the nerve, probe insertion is oriented parallel to the target nerve [47]. A diagnostic nerve block (e.g., with lidocaine 1–2% for rapid pain relief and bupivacaine 0.25–0.5% for longer-lasting analgesia, with optional steroids to reduce inflammation) is recommended prior to cryoneurolysis. The rationale for this two-step approach is to confirm the target nerve and also to familiarize the patient with expected post-procedural conditions [47]. If there is a positive response to the diagnostic block, i.e., pain alleviation, cryoneurolysis is performed in a subsequent intervention (Fig. 1). In patients with cancer-related pain syndromes, direct cryoneurolysis without a preliminary diagnostic block is deemed acceptable [47]. Advanced techniques such as selective MR neurography-guided nerve blocks may be helpful, especially in identifying neuropathies of the pudendal nerve, posterior femoral cutaneous nerve, lateral cutaneous nerve, and obturator nerve [54].

Again, there are currently no standardized protocols for cryoneurolysis. Both single and alternating freeze-thaw cycles are being employed (3–10 min freeze and 3–5 min thaw), respectively. For the duration of cycles, most experts rely on imaging visualization and validated protocols [47, 53]. Under- or over-ablation causing allodynia or pain exacerbation should be avoided. Further potential side effects (which are extremely rare) include bleeding, infection, frostbite, alopecia, and depigmentation. During freezing, temperatures between −80 °C and −20 °C should be maintained in order to prevent over- or underablation, and in order to induce a controlled Sunderland type II nerve injury. Potential transient sensory changes or weakness may occur when targeting mixed nerves. A structured clinical follow-up (e.g., at 1 week, 4 weeks, 3, 6, and 12 months) is strongly recommended to track pain relief duration and to adjust treatment protocols based on patient response [47]. Imaging is not

Table 2 Most important studies summarizing clinical evidence for cryoneurolysis

Target and clinical indication	Study design/description	Main findings	Reference
Peripheral nerves			
ON in occipital neuralgia, following a positive response to local anesthesia injection	Retrospective single-center study; 38 patients (20 unilateral, greater ON; 10 unilateral greater and lesser ON; 8 bilateral greater ON)	Average improvement of pain relief following cryoneurolysis 57.9% (mean duration 6.1 months overall); pain relief was even enhanced in patients reporting a pain relief of at least 75% following local anesthetic injections; adverse events: post-procedure neuritis ($n = 2$), hematoma ($n = 1$)	Kim et al 2015 [79]
Intercostal nerves prior to thoracotomy (in patients undergoing lobectomy or pneumectomy)	Prospective randomized single-center study; 90 patients ($n = 45$ epidural analgesia alone vs. $n = 45$ epidural analgesia + cryoneurolysis)	Cryoneurolysis combined with thoracic epidural analgesia was associated with earlier recovery in pulmonary function, less pain during movement, and a lower daily requirement for rescue analgesia 1 week after surgery; no significant decrease in incidence of long-term pain and numbness in patients undergoing cryoneurolysis	Yang et al 2004 [80]
Intercostal nerves prior to open thoracic or thoracoabdominal aneurysm repair	Retrospective, comparative single-center study; 117 patients (25 patients received cryoneurolysis, 92 patients did not)	Average use of post-surgical narcotics was significantly reduced in the cryoneurolysis group; no difference in post-operative strokes, paraplegia, pneumonia, and in-hospital mortality between the non-cryoneurolysis and cryoneurolysis group	Clemence et al 2020 [81]
Medial branch of lumbar facet joint in patients with chronic back pain	Prospective single-center study including 46 patients	At 6 weeks follow-up, 72% of patients were free of pain or had major improvement of lower back pain; 28% of patients had no or only little improvement	Birkenmaier et al 2007 [61]
Pudendal nerve in refractory pudendal neuralgia	Retrospective single-center study including 11 patients	Significant decrease of average level of pain at 24 h, 45 days, and 6 months post treatment; no procedure-related complications	Prologo et al 2014 [82]
Various nerve targets in patients with chronic refractory peripheral neuropathic pain	Prospective observational single-center study including 22 patients: ilioinguinal nerve ($n = 3$), posterior tibial nerve ($n = 4$), saphenous nerve ($n = 7$), gluteal nerve ($n = 1$), sural nerve ($n = 1$), geniculate nerve ($n = 1$), digital nerve ($n = 2$), plantar neuroma ($n = 3$)	Significant decrease in post-procedural pain scores, with moderate long-term relief at 12-month follow-up	Yoon et al 2016 [83]
Lateral femoral cutaneous nerve prior to skin graft harvesting in burn patients	Randomized controlled pilot study comparing cryoneurolysis (6 patients) vs. sham procedure (6 patients)	Patients in the cryoneurolysis group experienced lower average and maximum pain scores on post-operative days 1–4 and 7; patients in the cryoneurolysis group were less likely to report sleep disturbances for the first two nights following surgery	Finneran et al 2022 [88]

Table 2 continued

Target and clinical indication	Study design/description	Main findings	Reference
Infrapatellar branch of the saphenous nerve in patients with mild to moderate knee osteoarthritis	Sham-controlled, double-blind, multicenter study; 180 patients randomized 2:1 (cryoneurolysis vs. sham treatment)	Patients who received active treatment had significantly greater pain relief compared to sham treatment for up to 150 days; most expected side effects were mild in severity and resolved within 30 days; incidence of device- or procedure-related adverse events was similar in both groups	Radnovich et al 2017 [64]
Superficial genicular nerves in patients with osteoarthritis prior to total knee arthroplasty	Unblinded randomized controlled single-center study; comparison of cryoneurolysis (62 patients) vs. standard of care (62 patients)	Compared with the standard of care, patients undergoing cryoneurolysis had improved functional scores and improved pain scores at 72-h and 2-week follow-up	Mihalko et al 2021 [84]
Genicular nerves in patients with knee osteoarthritis	Systematic review and meta-analysis; comparison of RFA, cryoneurolysis, and sham treatment; inclusion of eight randomized controlled trials (627 patients)	Compared with sham treatment, ablative techniques significantly reduced pain at 12 weeks follow-up and improved functional scores; subgroup analysis favored conventional monopolar RFA over other modalities; no serious adverse events were reported	Barreto et al 2026 [85]
Sympathetic neurolysis (ganglia and plexus)			
Celiac plexus in patients with pancreatic adenocarcinoma	Prospective observational study of 9 patients; data so far only published as an abstract from the World Congress of Interventional Oncology	Five patients had permanent decrease or resolution of pain, including two patients with immediate post-procedure pain relief lasting up to 8 weeks; two patients did not experience pain relief following cryoablation; one patient experienced diarrhea immediately after cryoablation, which resolved spontaneously	Mortell et al 2017 [86]
Celiac plexus in patients with intractable abdominal pain due to cancer, pancreatitis, or a benign cause	Retrospective single-center study; cohort comparison of cryoneurolysis (39 patients) and ethanol injection (44 patients)	Mean follow-up 17.7 days; significantly higher incidence of gastrointestinal symptoms (nausea, vomiting, diarrhea) in the ethanol group compared with the cryoneurolysis group (27.3% vs. 10.3%); decrease of average pain score by 46.7% in the cryoneurolysis group vs. 26.6% in the ethanol group	Behbahan et al 2020 [68]
Splanchnic nerves in patients with pancreatic-cancer-related pain	Retrospective single-center study including 50 patients	Technical success 100%; clinically relevant pain reduction was achieved in 76% of patients; baseline mean self-reported pain (9.52/10) was reduced to 4.36, 4.23, and 4.43 at 1-, 6-, and 12-month follow-up; reduction of opioid analgesia was observed in 76% of patients; no severe complications	Chlorogiannis et al 2025 [87]
Neuroma			
Sensitized neuroma in patients post limb amputation	Prospective single-center study including 21 patients	Technical success 100%; 6 minor procedure-related complications; significant decrease of disability scores and pain intensity	Prologo et al 2017 [73]

Table 2 continued

Target and clinical indication	Study design/description	Main findings	Reference
Sensitized neuroma in patients post limb amputation	Prospective observational single-center study including 7 patients (8 neuromas)	Technical success 100%; no major complications; significant decrease of pain at the mean follow-up of 27 months; high patient satisfaction	von Falck et al 2022 [72]
Morton neuroma	Retrospective single-center study including 20 patients	Technical success 100%; significant decrease of mean pain score; high patient satisfaction; no instance of stump neuroma formation	Cazzato et al 2016 [74]
Morton neuroma	Retrospective single-center study including 54 patients (59 neuromas)	Technical success 98.1%; 3 minor complications; at 6 months follow-up, significantly decreased mean pain score; complete pain relief in 60.4%, partial pain relief in 28.3%, and no pain relief or increased pain in 11.3% of patients	Moulin et al 2024 [74]

ON occipital nerve, *RFA* radiofrequency ablation

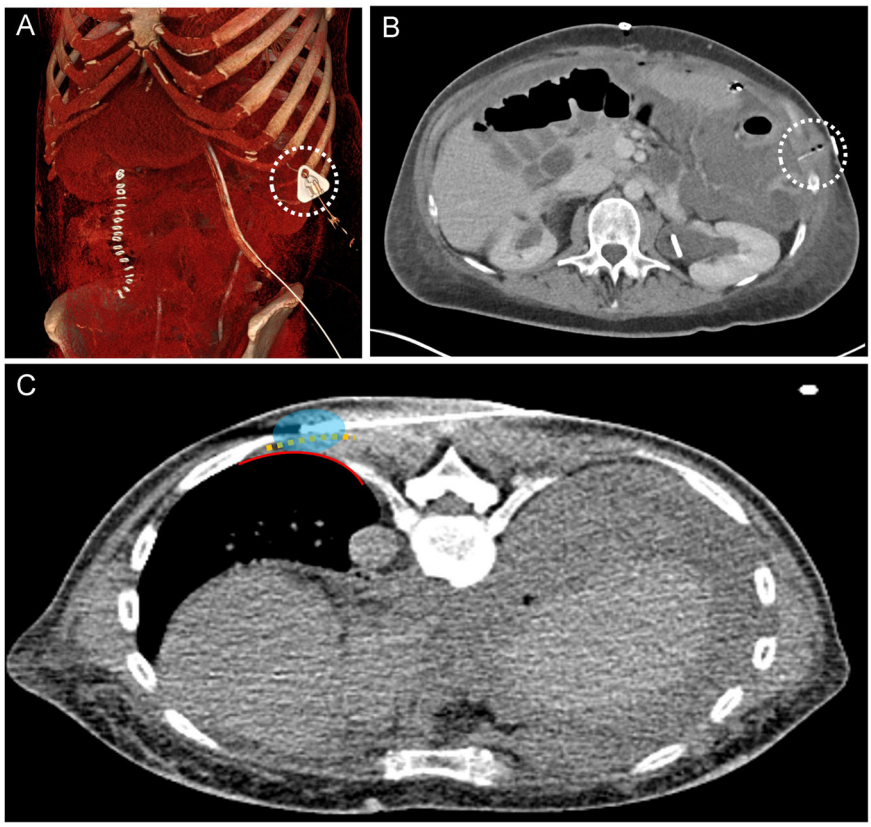


Fig. 4 Intercostal cryoneurolysis. 42-year-old female patient with recurrent pseudomyxoma peritonei. Following surgical gastrostomy placement via an intercostal approach, the patient suffered from intractable neuropathic pain caused by irritation of the nerve from the tube (circle in **A** and **B**). Following a diagnostic nerve block, a single cryoprobe was placed encompassing the 9th intercostal nerve (in **C**: projected course of nerve is displayed as dotted line; projected ice ball is marked in blue; pleura is denoted as red line), and cryoneurolysis was performed (two 6 min freeze cycles with intermittent 4 min of passive thawing) (**C**)



Fig. 5 Splanchnic nerve cryoneurolysis. 56-year-old male patient with intractable pain from pancreatic cancer. CT-guided placement of two cryoprobes via a posterolateral approach to the target region, which is located retrocrural at the anterolateral border of the T12 vertebral body (*marks the aorta). Following bilateral placement of the cryoprobes medially to the diaphragmatic crura (arrows), cryoneurolysis was performed (one 10-min freeze cycle followed by 4 min of passive and 1 min of active thawing)

required routinely to assess treatment response, but is recommended when complications are suspected or symptoms change unexpectedly. If cryoneurolysis is unsuccessful or leads to only partial pain alleviation, treatment may be repeated within 2–4 weeks [47]. Despite the increasing body of evidence on the use of cryoneurolysis [55], it has been proposed that future research should focus on randomized trials, standardized outcome reporting, predictors of success, and safety surveillance [8] (Table 2).

Peripheral nerves

It has been shown that cryoneurolysis may be highly effective for the treatment of acute or chronic pain (refractory to conservative therapies) whenever peripheral nerve targets can be identified. Even though current evidence is mostly based on single-center case series and retrospective studies, results are very encouraging and reflect a broad spectrum of potential indications in neoplastic and non-neoplastic conditions. As such, cryoneurolysis has been associated with various degrees of decreased pain and narcotic utilization, avoidance of invasive surgery, and cost-effectiveness in selected patients. Increasing evidence supports the application of cryoneurolysis for pain management in a variety of neuralgias and conditions [53]. An enumeration and in-depth discussion of all potential indications lies beyond the scope of this review. This includes for example pain related to abdominal, pelvic or thoracic neoplasms, but similarly pain related to benign substrate such as occipital neuralgia [56], inguinodynia [57], chronic orchialgia, pain related to burns or tissue injury [58], post-herpetic neuralgia [59], various peripheral nerve neuralgias as well as biomechanical syndromes including lumbar and cervical facet joint osteoarthritis [60, 61]. Studies directly comparing the effectiveness of steroid injections and cryotherapy for the treatment of facet joint pain are currently lacking. However, it has been suggested that patients may benefit from facet joint neurolysis, specifically by RFA or cryoablation, following a positive diagnostic facet joint block [62]. Furthermore, it has been demonstrated that patients undergoing surgical procedures, including total knee arthroplasties, shoulder arthroplasties, thoracotomies, or mastectomies, benefit from cryoneurolysis performed prior to the surgery in terms of greater post-surgery pain relief [63]. Cryoneurolysis may be an effective alternative to conservative/medical treatment in patients with osteoarthritis. A randomized, double-blind, multicenter study, for example, included 180 patients with knee osteoarthritis [64]. Patients undergoing cryoneurolysis reported statistically significant improvement of symptoms for up to 150 days compared to patients undergoing sham treatment, with no serious adverse events being observed [64]. Similarly, cryoneurolysis should be considered in selected patients for treatment of post-surgical pain syndromes (Fig. 4).

Sympathetic neurolysis (ganglia and plexus)

Especially in cancer patients, depending on tumor location and spread specific target sites for pain palliation include splanchnic and celiac plexus, stellate ganglion, thoracic plexus, lumbar plexus, superior hypogastric plexus and ganglion impar [65, 66] (Fig. 5). The most common targets for sympathetic neurolysis are the splanchnic and celiac plexus, especially in patients with pancreatic cancer, hepatobiliary or upper gastrointestinal cancers, as well as retroperitoneal lymph node metastases [67]. Compared with chemical neurolysis using ethanol, cryoneurolysis of the celiac plexus, for example, has been shown to be associated with a lower incidence of diarrhea and fewer gastrointestinal side effects [68]. In addition, the controllability of cryoneurolysis is superior to ethanol distribution and is not limited by bulky tumors. Stellate ganglion neurolysis may be considered in patients with neck, facial, or Pancoast tumors, or breast cancer patients suffering from lymphedema [66]. Superior hypogastric plexus and ganglion impar neurolysis have been successfully used in different visceral and perineal pelvic cancers [69].

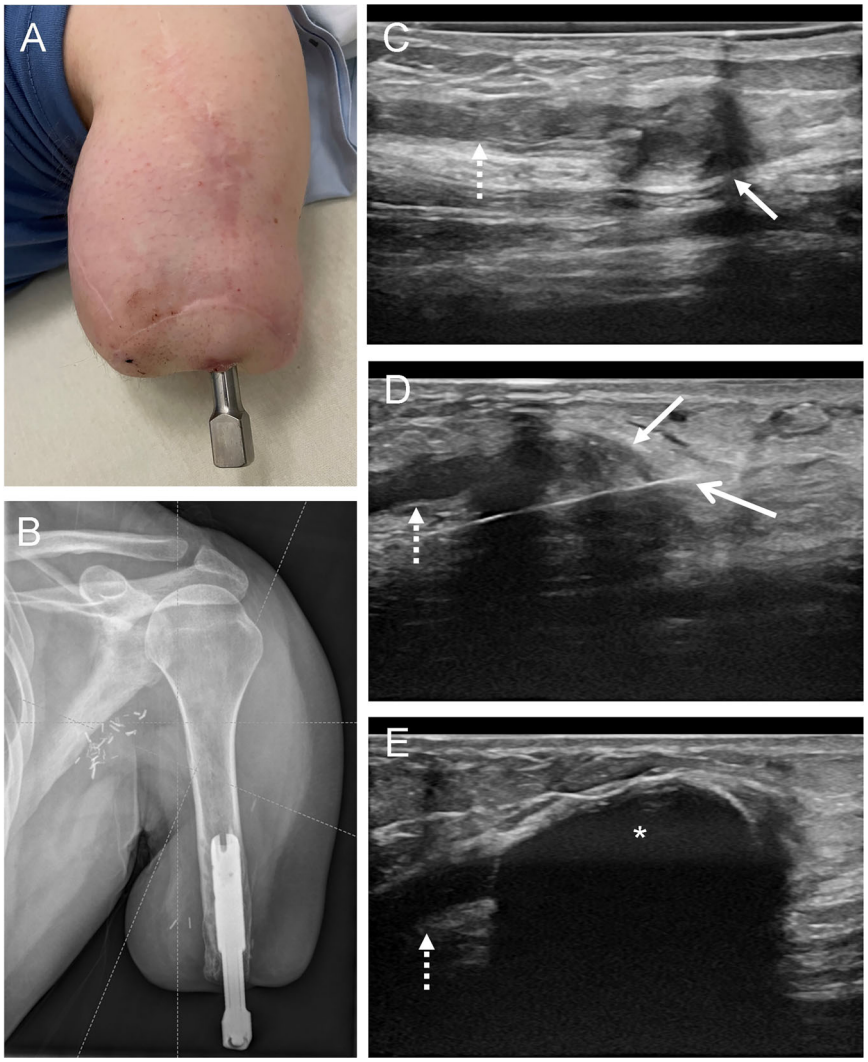


Fig. 6 Ultrasound-guided cryoneurolysis of a post-amputation stump neuroma. 38-year-old male patient, 4 years post upper limb amputation following a motor vehicle accident. The patient suffered from massive neuropathic pain, including hyperpathia, despite pain medication, and was not able to wear his endo-exo prosthesis for more than a couple of minutes without pain. Clinically (**A**) and at radiography (**B**), the stump appeared unremarkable. Ultrasound (**C**) identified an enlarged cutaneous brachii medialis nerve (dashed arrow) along with a sensitized neuroma measuring 12 mm (arrow), which could be triggered by pressure. Following a positive diagnostic block (not shown), ultrasound-guided placement of a single cryoprobe (open arrow in **D**) and cryoneurolysis was performed (two cycles of 6-min freezing separated by a 4-min thaw cycle). The evolving ice ball (* in **E**) encompassing the neuroma was monitored continuously. Two hours after the procedure, the patient was able to wear his prosthesis without any residual pain

Sensitized neuroma

Neuropathic pain is a common complaint in patients following upper or lower limb amputation, with up to 85% of amputees complaining of pain perceived in the stump and/or phantom limb [70]. Neuroma formation is a typical response following nerve injury, characterized by a benign conglomerate of cells and axons embedded in a dense fibrotic matrix [71]. Neuropathic pain syndromes are very complex and multidimensional. However, in case of post-amputation pain, a sensitized neuroma may be identified as a potential cause in nearly 50% of patients, causing debilitating pain and significant impairment in quality of life [70, 72]. Cryoneurolysis of a sensitized stump neuroma in patients with therapy-refractory pain has been shown to be highly effective and safe with very encouraging short- and long-term results (Fig. 6). Marked pain alleviation is typically achieved within hours of treatment, going hand in hand with a very high patient satisfaction [72, 73]. Similarly, cryoneurolysis of Morton neuroma has been reported with technical success rates of

up to 100% [74]. In a recent study including 54 patients, 60.4% of patients reported complete pain relief at 6 months, and 11.3% of patients reported partial pain relief, respectively [75]. Of course, these promising results need to be confirmed at long-term follow-up.

Conclusion

Interventional radiology has become a cornerstone of modern pain therapy and should be an integral component of interdisciplinary treatment concepts. As such, the role of cryoablation and cryoneurolysis for treatment of acute and chronic pain syndromes, in patients with cancer and non-cancer-related pain syndromes, is continuously evolving with a growing body of evidence. Appropriate patient selection and education, with regard to outcome expectations, are the first steps to guiding treatment success. Identification of the specific treatment target, choice of technique, and precise treatment execution are further key determinants and indispensable to building a successful interventional pain practice.

Abbreviations

MWA	Microwave ablation
RFA	Radiofrequency ablation
VAS	Visual analog scale

Funding

Open Access funding enabled and organized by Projekt DEAL.

Compliance with ethical standards

Guarantor

The scientific guarantor of this publication is Kristina Ringe.

Conflict of interest

The authors of this manuscript declare relationships with the following companies: Daniel M. Düx: none. Dimitrios Filippiadis: Honoraria from Boston Scientific and Siemens Healthineers; not in relation to this work. Frank Wacker: Institutional research grants from Siemens Healthineers, Promedicus, Delcath; travel support from Philips Healthcare; not in relation to this work. Kristina I. Ringe: Honoraria from Boston Scientific and Siemens Healthineers; not in relation to this work.

Statistics and biometry

No complex statistical methods were necessary for this paper.

Informed consent

Written informed consent was not required for this study because this is a review article.

Ethical approval

Institutional Review Board approval was not required because this is a review article.

Study subjects or cohorts overlap

Not applicable.

Methodology

- Invited review article

Received: 9 June 2026 Revised: 30 September 2026 Accepted: 9 October 2026
Published online: 06 November 2026

References

1. Treede RD, Rief W, Barke A et al (2015) A classification of chronic pain for ICD-11. Pain 156:1003–1007
2. van den Beuken-van Everdingen MH, Hochstenbach LM, Joosten EA, Tjan-Heijnen VC, Janssen DJ (2016) Update on prevalence of pain in patients with cancer: systematic review and meta-analysis. J Pain Symptom Manage 51:1070–1090.e1079
3. Dahlhamer J, Lucas J, Zelaya C et al (2018) Prevalence of chronic pain and high-impact chronic pain among adults—United States, 2016. Morb Mortal Wkly Rep 67:1001–1006
4. Garnon J, Cazzato RL, Caudrelier J et al (2019) Adjunctive thermoprotection during percutaneous thermal ablation procedures: review of current techniques. Cardiovasc Interv Radiol 42:344–357
5. Trescot AM (2003) Cryoanalgesia in interventional pain management. Pain Physician 6:345–360
6. de Baere T, Woodrum D, Tselikas L et al (2021) The ECLIPSE study: efficacy of cryoablation on metastatic lung tumors with a 5-year follow-up. J Thorac Oncol 16:1840–1849
7. Lloyd JW, Barnard JD, Glynn CJ (1976) Cryoanalgesia. A new approach to pain relief. Lancet 2:932–934
8. Cornelis FH, Resnick NJ, Barral M et al (2026) Interventional management of pain with cryoneurolysis: expert consensus recommendations for standardizing patient selection and techniques based on Delphi process. Cardiovasc Interv Radiol 49:585–593
9. Gravel G, Nobileau A, Guth A, Mellot F, Roussel A (2025) Interventional radiology management of bone metastasis pain: strategies and techniques. Cardiovasc Interv Radiol 48:907–918
10. Jitsinthunun T, Li C, Ng TK, Zinboonyahgoon N (2025) Pulsed radiofrequency treatment: evidence for and applications in chronic pain. Pain Physician 28:467–481
11. Georgiades CS, Marx JK (2011) Cryoablation: mechanism of action and devices. In: Hong KG, Georgiades CS (eds) Percutaneous tumor ablation: strategies and techniques. Thieme, New York, pp 15–26
12. Baust JG, Gage AA, Bjerklund Johansen TE, Baust JM (2014) Mechanisms of cryoablation: clinical consequences on malignant tumors. Cryobiology 68:1–11
13. Gage AA, Baust J (1998) Mechanisms of tissue injury in cryosurgery. Cryobiology 37:171–186
14. Klossner DP, Robilotto AT, Clarke DM et al (2007) Cryosurgical technique: assessment of the fundamental variables using human prostate cancer model systems. Cryobiology 55:189–199
15. Erinjeri JP, Clark TW (2010) Cryoablation: mechanism of action and devices. J Vasc Interv Radiol 21:S187–S191
16. Baust JG, Gage AA (2005) The molecular basis of cryosurgery. BJU Int 95:1187–1191
17. Baust JG, Gage AA, Robilottto AT, Baust JM (2009) The pathophysiology of thermoablation: optimizing cryoablation. Curr Opin Urol 19:127–132
18. Aarts BM, Klompenhouwer EG, Rice SL et al (2019) Cryoablation and immunotherapy: an overview of evidence on its synergy. Insights Imaging 10:53
19. Kwak K, Yu B, Lewandowski RJ, Kim DH (2022) Recent progress in cryoablation cancer therapy and nanoparticles mediated cryoablation. Theranostics 12:2175–2204
20. Sunderland S (1951) A classification of peripheral nerve injuries producing loss of function. Brain 74:491–516
21. Zhou L, Shao Z, Ou S (2003) Cryoanalgesia: electrophysiology at different temperatures. Cryobiology 46:26–32
22. Wagner R, DeLeo JA, Heckman HM, Myers RR (1995) Peripheral nerve pathology following sciatic cryoneurolysis: relationship to neuropathic behaviors in the rat. Exp Neurol 133:256–264
23. Ilfeld BM, Preciado J, Trescot AM (2016) Novel cryoneurolysis device for the treatment of sensory and motor peripheral nerves. Expert Rev Med Devices 13:713–725

24. Prologo JD, Edalat F, Moussa M (2020) Interventional cryoneurolysis: an illustrative approach. Tech Vasc Interv Radiol 23:100698
25. Hadi M, Walker C, Desborough M et al (2021) CIRSE standards of practice on peri-operative anticoagulation management during interventional radiology procedures. Cardiovasc Interv Radiol 44:523–536
26. Malhotra A, Mackey S (2012) Outcomes in pain medicine: a brief review. Pain Ther 1:5
27. Le Corroller T, Arrigoni F, Guenoun D, Rochwerger A, Champsaur P, Mattei JC (2026) Percutaneous cryoablation for the management of bone and soft tissue tumors. Semin Musculoskelet Radiol 30:61–74
28. Khanmohammadi S, Noroozi A, Yekaninejad MS, Rezaei N (2023) Cryoablation for the palliation of painful bone metastasis: a systematic review. Cardiovasc Interv Radiol 46:1469–1482
29. Alaqeel M, Alnajres A, Mulia Y et al (2025) Radiofrequency ablation and cryoablation in treating painful bone metastasis: a comprehensive systematic review and separate single-arm meta-analysis. Orthop Rev 17:150315
30. Thacker PG, Callstrom MR, Curry TB et al (2011) Palliation of painful metastatic disease involving bone with imaging-guided treatment: comparison of patients' immediate response to radiofrequency ablation and cryoablation. AJR Am J Roentgenol 197:510–515
31. Le Corroller T, Arrigoni F (2025) ESR Essentials: percutaneous bone consolidation-practice recommendations by the European Society of Musculoskeletal Radiology. Eur Radiol 35:5369–5380
32. Mesny E, Martz N, Stacoffe N et al (2025) State-of-the-art of multidisciplinary approach of bone metastasis-directed therapy: review and challenging questions for preparation of a GEMO practice guidelines. Cancer Metastasis Rev 44:45
33. Di Staso M, Gravina GL, Zugaro L et al (2015) Treatment of solitary painful osseous metastases with radiotherapy, cryoablation or combined therapy: propensity matching analysis in 175 patients. PLoS One 10:e0129021
34. Kransdorf MJ, Stull MA, Gilkey FW, Moser RP Jr (1991) Osteoid osteoma. Radiographics 11:671–696
35. Motamedi D, Learch TJ, Ishimitsu DN et al (2009) Thermal ablation of osteoid osteoma: overview and step-by-step guide. Radiographics 29:2127–2141
36. Rosenthal DI, Alexander A, Rosenberg AE, Springfield D (1992) Ablation of osteoid osteomas with a percutaneously placed electrode: a new procedure. Radiology 183:29–33
37. Ringe KI, Panzica M, von Falck C (2016) Thermoablation of bone tumors. Rofo 188:539–550
38. Lindquester WS, Crowley J, Hawkins CM (2020) Percutaneous thermal ablation for treatment of osteoid osteoma: a systematic review and analysis. Skelet Radiol 49:1403–1411
39. Shanmugasundaram S, Nadkarni S, Kumar A, Shukla PA (2021) Percutaneous ablative therapies for the management of osteoid osteomas: a systematic review and meta-analysis. Cardiovasc Interv Radiol 44:739–749
40. Le Corroller T, Vives T, Mattei JC et al (2022) Osteoid osteoma: percutaneous CT-guided cryoablation is a safe, effective, and durable treatment option in adults. Radiology 302:392–399
41. Pusceddu C, Vergantino E, Santucci D et al (2023) Percutaneous cryoablation under conscious sedation: a safe, effective and painless option for the treatment of pediatric osteoid osteoma. J Clin Med 12:6889
42. Coupal TM, Mallinson PI, Munk PL, Liu D, Clarkson P, Ouellette H (2014) CT-guided percutaneous cryoablation for osteoid osteoma: initial experience in adults. AJR Am J Roentgenol 202:1136–1139
43. Santiago E, Pauly V, Brun G, Guenoun D, Champsaur P, Le Corroller T (2018) Percutaneous cryoablation for the treatment of osteoid osteoma in the adult population. Eur Radiol 28:2336–2344
44. Cazzato RL, Auloge P, Dalili D et al (2019) Percutaneous image-guided cryoablation of osteoblastoma. AJR Am J Roentgenol 213:1157–1162
45. Serrano E, Zarco F, Gill AE et al (2021) Percutaneous cryoablation of chondroblastoma and osteoblastoma in pediatric patients. Insights Imaging 12:106
46. Gowda PC, Dunlap RH, Ahlawat S, Gross JM, Morris CD, Lyons GR (2023) Recurrent chondromyxoid fibroma of the distal femur treated with percutaneous cryoablation. Skelet Radiol 52:2497–2501
47. Cornelis FH, Marin F, Labreze C et al (2017) Percutaneous cryoablation of symptomatic venous malformations as a second-line therapeutic option: a five-year single institution experience. Eur Radiol 27:5015–5023
48. Kurtz JE, Buy X, Deschamps F et al (2021) CRYODESMO-O1: a prospective, open phase II study of cryoablation in desmoid tumour patients progressing after medical treatment. Eur J Cancer 143:78–87
49. Cazzato RL, Gantzer J, de Marini P et al (2022) Sporadic desmoid tumours: systematic review with reflection on the role of cryoablation. Cardiovasc Interv Radiol 45:613–621
50. Maillot J, Brun JL, Dubuisson V, Bazot M, Grenier N, Cornelis FH (2017) Mid-term outcomes after percutaneous cryoablation of symptomatic abdominal wall endometriosis: comparison with surgery alone in a single institution. Eur Radiol 27:4298–4306
51. Dibble EH, D'Amico KC, Bandera CA, Littrup PJ (2017) Cryoablation of abdominal wall endometriosis: a minimally invasive treatment. AJR Am J Roentgenol 209:690–696
52. Thompson SM, Schmitz JJ, Schmit GD, Callstrom MR, Kurup AN (2017) Image-guided thermal ablative therapies in the treatment of sarcoma. Curr Treat Options Oncol 18:25
53. Filippiadis D, Efthymiou E, Tsochatzis A, Kelekis A, Prologo JD (2021) Percutaneous cryoanalgesia for pain palliation: current status and future trends. Diagn Interv Imaging 102:273–278
54. Fritz J, Dellon AL, Williams EH, Belzberg AJ, Carrino JA (2015) 3-Tesla high-field magnetic resonance neurography for guiding nerve blocks and its role in pain management. Magn Reson Imaging Clin N Am 23:533–545
55. Cornelis FH, Bokhari R, Geevarghese R, Filippiadis DK (2025) Cryoneurolysis: parallel growth in PubMed publications and Google trends analysis reveals exponential interest. Cardiovasc Interv Radiol 48:1188–1192
56. Kastler A, Onana Y, Comte A, Attye A, Lajoie JL, Kastler B (2015) A simplified CT-guided approach for greater occipital nerve infiltration in the management of occipital neuralgia. Eur Radiol 25:2512–2518
57. Campos NA, Chiles JH, Plunkett AR (2009) Ultrasound-guided cryoablation of genitofemoral nerve for chronic inguinal pain. Pain Physician 12:997–1000
58. Finneran JJ, Swisher MW, Gabriel RA et al (2020) Ultrasound-guided lateral femoral cutaneous nerve cryoneurolysis for analgesia in patients with burns. J Burn Care Res 41:224–227
59. Weber G, Saad K, Awad M, Wong TH (2019) Case report of cryoneurolysis for the treatment of refractory intercostobrachial neuralgia with postherpetic neuralgia. Local Reg Anesth 12:103–107
60. Wolter T, Kleinmann B, Knoeller S (2018) Cryoneurolysis for the treatment of cervical facet joint syndrome: a technical note. J Pain Res 11:1165–1169
61. Birkenmaier C, Veihelmann A, Trouillier H et al (2007) Percutaneous cryodenervation of lumbar facet joints: a prospective clinical trial. Int Orthop 31:525–530
62. Perolat R, Kastler A, Nicot B et al (2018) Facet joint syndrome: from diagnosis to interventional management. Insights Imaging 9:773–789
63. Biel E, Aroke EN, Maye J, Zhang SJ (2023) The applications of cryoneurolysis for acute and chronic pain management. Pain Pract 23:204–215
64. Radnovich R, Scott D, Patel AT et al (2017) Cryoneurolysis to treat the pain and symptoms of knee osteoarthritis: a multicenter, randomized, double-blind, sham-controlled trial. Osteoarthritis Cartilage 25:1247–1256
65. Filippiadis DK, Tselikas L, Tsitskari M, Kelekis A, de Baere T, Ryan AG (2019) Percutaneous neurolysis for pain management in oncological patients. Cardiovasc Interv Radiol 42:791–799
66. Heptonstall N, Scott-Warren J, Berman R, Filippiadis D, Bell J (2023) Role of interventional radiology in pain management in oncology patients. Clin Radiol 78:245–253
67. Chary A, Edalat F (2022) Celiac plexus cryoneurolysis. Semin Interv Radiol 39:138–141
68. Behbahani K, Chary A, Patel S, Mitchell JW, Fleishon H, Prologo JD (2020) Percutaneous CT-guided cryoablation of the celiac plexus: a retrospective cohort comparison with ethanol. J Vasc Interv Radiol 31:1216–1220
69. Ahmed DG, Mohamed MF, Mohamed SA (2015) Superior hypogastric plexus combined with ganglion impar neurolytic blocks for pelvic and/or perineal cancer pain relief. Pain Physician 18:E49–E56
70. Buchheit T, Van de Ven T, Hsia HL et al (2016) Pain phenotypes and associated clinical risk factors following traumatic amputation: results from veterans integrated pain evaluation research (VIPER). Pain Med 17:149–161
71. Oliveira KMC, Pindur L, Han Z, Bhavsar MB, Barker JH, Leppik L (2018) Time course of traumatic neuroma development. PLoS One 13:e0200548

72. von Falck C, Orgel M, Wacker F, Aschoff HH, Krettek C, Ringe KI (2022) Icing the pain-ultrasound-guided cryoablation of symptomatic post-amputation stump neuroma. Cardiovasc Interv Radiol 45:223–227
73. Prologo JD, Gilliland CA, Miller M et al (2017) Percutaneous image-guided cryoablation for the treatment of phantom limb pain in amputees: a pilot study. J Vasc Interv Radiol 28:24–34.e24
74. Cazzato RL, Garnon J, Ramamurthy N et al (2016) Percutaneous MR-guided cryoablation of Morton's neuroma: rationale and technical details after the first 20 patients. Cardiovasc Interv Radiol 39:1491–1498
75. Moulin B, Angelopoulos G, Sarrazin JL et al (2024) Safety and efficacy of percutaneous morton neuroma cryoneurolysis under ultrasound guidance. Cardiovasc Interv Radiol 47:795–800
76. Callstrom MR, Dupuy DE, Solomon SB et al (2013) Percutaneous image-guided cryoablation of painful metastases involving bone: multicenter trial. Cancer 119:1033–1041
77. Deschamps F, Farouil G, Ternes N et al (2014) Thermal ablation techniques: a curative treatment of bone metastases in selected patients? Eur Radiol 24:1971–1980
78. Thompson RE, Ajrawat H, Solomon AJ et al (2026) Cryoablation versus radiofrequency ablation for the treatment of osteoid osteoma. J Vasc Interv Radiol 37:108848
79. Kim CH, Hu W, Gao J, Dragan K, Whealton T, Julian C (2015) Cryoablation for the treatment of occipital neuralgia. Pain Physician 18:E363–E368
80. Yang MK, Cho CH, Kim YC (2004) The effects of cryoanalgesia combined with thoracic epidural analgesia in patients undergoing thoracotomy. Anaesthesia 59:1073–1077
81. Clemence J Jr, Malik A, Farhat L et al (2020) Cryoablation of intercostal nerves decreased narcotic usage after thoracic or thoracoabdominal aortic aneurysm repair. Semin Thorac Cardiovasc Surg 32:404–412
82. Prologo JD, Lin RC, Williams R, Corn D (2015) Percutaneous CT-guided cryoablation for the treatment of refractory pudendal neuralgia. Skelet Radiol 44:709–714
83. Yoon JH, Grechushkin V, Chaudhry A, Bhattacharji P, Durkin B, Moore W (2016) Cryoneurolysis in patients with refractory chronic peripheral neuropathic pain. J Vasc Interv Radiol 27:239–243
84. Mihalko WM, Kerkhof AL, Ford MC, Crockarell JR Jr, Harkess JW, Guyton JL (2021) Cryoneurolysis before total knee arthroplasty in patients with severe osteoarthritis for reduction of postoperative pain and opioid use in a single-center randomized controlled trial. J Arthroplasty 36:1590–1598
85. Barreto RB, Barreto AJ, do Nascimento ALM, Carvalho DMB, Queiroz B (2026) Efficacy and safety of genicular nerve ablation techniques for knee osteoarthritis: a systematic review and meta-analysis of sham-controlled randomized trials. Pain Med 27:449–461
86. Mortell K, Yamohammadi H, Brocone M, Haaga J, Nakamoto D (2014) Percutaneous CT-guided cryoablation of the celiac plexus in the palliative treatment of pancreatic cancer. J Vasc Interv Radiol 25:817.e15
87. Chlorogiannis DD, Kotsantis I, Economopoulou P et al (2025) Percutaneous cryoneurolysis of splanchnic nerves for pain palliation in patients with pancreatic cancer: a single-center experience. Cardiovasc Interv Radiol 48:1438–1446
88. Finneran JJ, Schaar AN, Swisher MW et al (2022) Percutaneous cryoneurolysis of the lateral femoral cutaneous nerve for analgesia following skin grafting: a randomized, controlled pilot study. Reg Anesth Pain Med 47:60–61

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.